

Number Sense

Integers · BEDMAS · Exponents · Fractions · Roots · Percent · Ratios

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise integers, exponent laws, fractions, roots, percents and ratios independently.

Show your full working in the space provided under each question. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the Answer Key.

Level 1 — Fluency

1. Evaluate: $(-11) + (-8) - (-5)$. **[Integers]**

2. Evaluate: $45 - 5 \times 3 + 2^2$. **[BEDMAS]**

3. Write as a single power: $9^4 \times 9^3$. **[Exponents]**

4. Evaluate in lowest terms: $7/10 - 1/5$. **[Fractions]**

5. Write 0.35 as a fraction in lowest terms and as a percent. **[Decimals/Percent]**

6. Evaluate: $\sqrt{144}$, and state whether it is rational. **[Roots]**

Level 2 — Application

7. Evaluate in lowest terms: $\frac{3}{8} \times \frac{4}{9} \div \frac{1}{3}$. **[Fractions]**

8. Simplify to a single power and evaluate: $(4^2)^2 \div 4^2$. **[Exponents]**

9. A \$120 coat is discounted 35%. Find the sale price. **[Percent (word)]**

10. A recipe uses sugar to flour in the ratio 2 : 5. If 350 g of flour is used, how much sugar is needed? **[Ratios (word)]**
11. Order from least to greatest: -0.3, $-\frac{1}{2}$, 0.25, $-\frac{2}{5}$. **[Rational numbers]**
12. Between which two consecutive whole numbers does $\sqrt{85}$ lie? Justify. **[Roots (estimate)]**

Level 3 — Thinking

13. After a 25% discount, a bike sells for \$180. Find the original price and check. **[Percent (work back)]**

14. Priya wrote $5^2 \times 5^3 = 25^5$. Explain her error and give the correct answer. **[Find the error]**
15. A 600 mL drink is 30% juice. How much pure juice must be added to make it 40% juice? **[Mixture]**
16. A runner covers 12 km in 1 h, then 6 km in 0.5 h. Find the average speed for the whole run and explain why it is not the mean of the two speeds. **[Rate reasoning]**

17. Evaluate in lowest terms: $(\frac{3}{4} - \frac{1}{2}) \div (\frac{2}{3} + \frac{1}{6})$. **[Fractions multi-step]**
18. On a map, 2 cm represents 15 km. Two towns are 9 cm apart on the map. Find the real distance. **[Proportion (scale)]**